

Background and Objectives

- Categorical semantics for finite-dimensional quantum theory is given by dagger compact closed categories.
- However it is a well known no-go theorem that models for infinite dimensional theory can't be compact closed.
- There are competing theories of dagger categories and mixed unitary categories to give semantics for the infinite-dimensional case. We aim to further study and develop the latter.
- In particular, we study one particular mathematical model of Lefschetz spaces, because of its interesting properties.

Motivation for a non-stationary dagger

Linear Logic Fragment	Categorical Semantics
MLL $(\otimes, \top), (\oplus, \perp)$	Linearly Distributive Categories (LDC)
MLL with negation	*-autonomous categories
Compact MLL $(\otimes = \oplus, \top = \perp)$	Monoidal Categories
Compact MLL with negation	Compact Closed Categories

Table 1. Linear Logic Perspective

Involutive and Unitary Categories

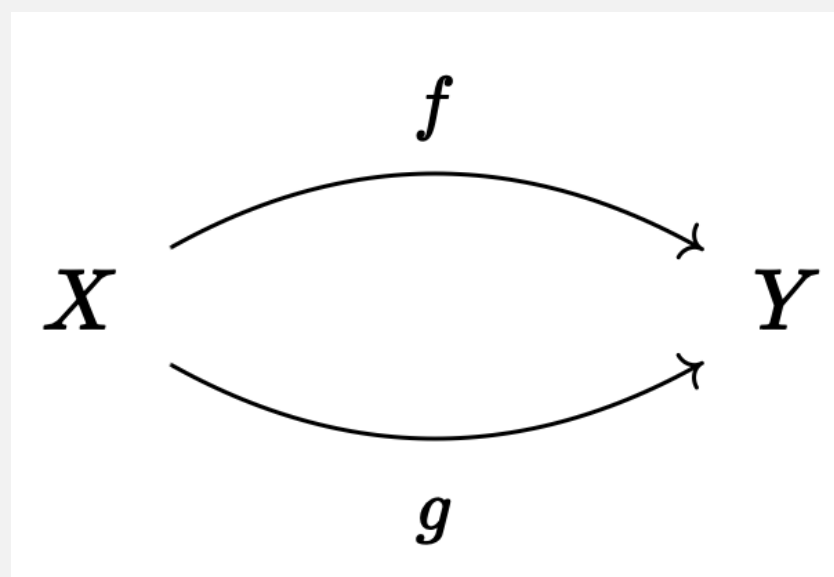
- In a $\dagger$ -category a map $f : A \rightarrow B$ is said to be unitary iff $f^\dagger \circ f = Id_A$ and $f \circ f^\dagger = Id_B$.
- However in an LDC, the dagger can't be made stationary on objects, so we aim to define involutive categories and unitary categories. We do so *independent* of the context of a LDC to focus on the core structure.
- A category $\mathbb{X}$ is said to be an involutive category if it is equipped with a contravariant endofunctor $()^\dagger : \mathbb{X}^{op} \rightarrow \mathbb{X}$ and a natural isomorphism $\iota : X \rightarrow X^{\dagger\dagger}$ for each $X \in \mathbb{X}_{obj}$.
- A functor $F : (\mathbb{X}, ()^\dagger, \iota) \rightarrow (\mathbb{Y}, ()^\ddagger, \iota')$ is said to be involutive if it is equipped with a preselector natural isomorphism $\gamma : F(X^\dagger) \rightarrow F(X)^\ddagger$ making the following diagrams commute:

$$\begin{array}{ccc}
 F(X) & \xrightarrow{F(\iota_X)} & F(X^{\dagger\dagger}) \\
 \downarrow \iota_{F(X)} & & \downarrow \gamma_{X^\dagger} \\
 F(X)^{\dagger\dagger} & \xrightarrow{\gamma_X^\ddagger} & F(X^\dagger)^\ddagger
 \end{array}$$

- A $\dagger$ -category is then when the dagger functor is stationary on objects and ι is the identity natural isomorphism.
- An involutive category is said to have the unitary structure if there is an essentially small class of objects $\mathbb{U} \subseteq Obj(\mathbb{X})$ which is closed to involution and such that for every $U \in \mathbb{U}$, there is an isomorphism, called its **unitary structure map**, such that $\varphi_{A^\dagger} = (\varphi_A^{-1})^\dagger$ and $\varphi_A \varphi_{A^\dagger} = \iota$
- A functor of unitary categories is a involutive functor that in addition preserves the unitary structure i.e. $\gamma_X \circ F(\varphi_X) = \varphi_{F(X)}$

$\dagger$ - Binary idempotents

- A binary idempotent in a category consists of two objects and opposing maps such that $f g f = f$ and $g f g = g$.



- A $\dagger$ -binary idempotent in an involutive category is a binary idempotent such that $f = \iota f^\dagger$ and $g^\dagger = g \iota$
- A morphism of $(\dagger)$ -binary idempotents is a pair of maps $(h, h') : (f, g) \rightarrow (f', g')$ such that $h = f h' g'$ and $h' = g h f'$

Main results

- We consider the category $\mathbf{BSplit}(\mathbb{X})$ where objects are binary idempotents and prove that it is complete with respect to splitting of idempotents. Moreover there is a full and faithful embedding of $\mathbb{X}$ into $\mathbf{BSplit}(\mathbb{X})$.
- If $\mathbb{X}$ is involutive, then $\mathbf{BSplit}(\mathbb{X})$ is involutive. Moreover, if we consider the full subcategory of $\dagger$ -binary idempotents is $\mathbb{X} \dagger = \mathbf{BSplit}(\mathbb{X})$, then it is involution closed.
- When $\mathbb{X}$ is involutive, then $\dagger = \mathbf{BSplit}(\mathbb{X})$ is a *unitary* category.
- An involutive functor $(F, \gamma) : (\mathbb{X}, \dagger, \iota) \rightarrow (\mathbb{Y}, \ddagger, \iota')$ induces a unitary functor between $\dagger = \mathbf{BSplit}(\mathbb{X})$ and $\dagger = \mathbf{BSplit}(\mathbb{Y})$.
- There is an equivalence between unitary categories and $\dagger$ categories.

Inner Product Structure

- An **inner product category** is an involutive category which has an **inner product combinator**:

$$\langle f | g \rangle : A \rightarrow A^\dagger f, g : A \rightarrow X$$

satisfying various coherence conditions.

- **Theorem:** An involutive category has an inner product combinator iff it has a unitary structure. If the unitary structure is trivial, it reduces to the usual notion of inner products.

Finiteness Spaces and Lefschetz Spaces

- A finiteness space $(X, \mathcal{U})$ is a set along with a family of subsets such that it is a fix point of the Galois adjunction taking a family to its finitary complement.
- A Lefschetz Space is a topological vector space such that it has an open neighbourhood basis around the origin of open *linear* subspaces.
- We do work with a different notion of convergence in Lefschetz spaces and thus with a different notion of basis.
- **Theorem:** Reflexive Lefschetz spaces $(V \cong V^{**})$ with a chosen basis $\mathbf{RLeft}_B$ and morphisms as continuous linear maps forms a Mixed Unitary Category, where the unitary core $(\otimes \cong \oplus, V \cong V^\dagger)$ is given by finite-dimensional spaces and unitary isomorphisms are given by maps taking the *chosen* basis elements to the chosen basis elements.

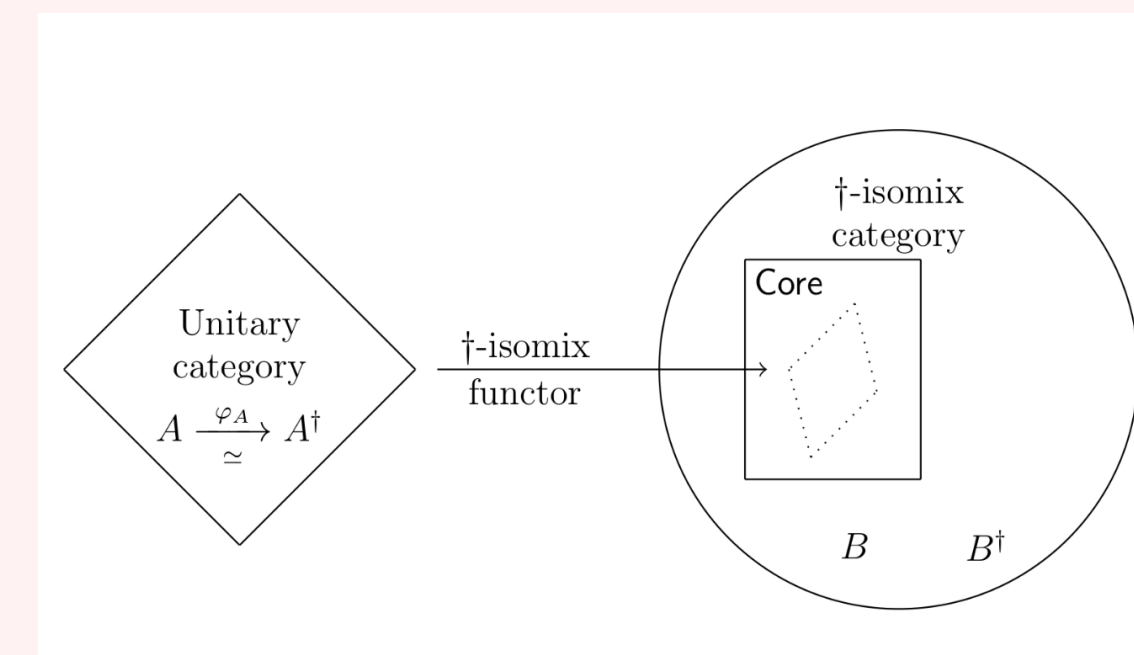


Figure 1. Idea of a MUC

- We show that there is a lax-functor from $\mathbf{RLeft}_B$ to the category of finiteness spaces and relations. This functor captures the support of convergent nets in Lefschetz spaces over the given basis.
- When restricted to the subcategory with positive maps, this lax functor turns into a strict monoidal functor.
- Every Reflexive Lefschetz space's topology arises as a finiteness topology, thus we have some classification theorems of its topology.

Open questions

- Is there a *canonical* unitary structure which gives rise to a global inner product?
- In finite-dimensional case, observables correspond to a choice of a $\dagger$ -Frobenius algebra, in arbitrary dimension does this correspondence carry to $\dagger$ -linear monoids?
- Can MUC give a semantics for quantum concurrency?

Main References

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- [3] Thomas Ehrhard. Finiteness spaces. *Mathematical Structures in Computer Science*, 15(4):615–646, 7 2005.